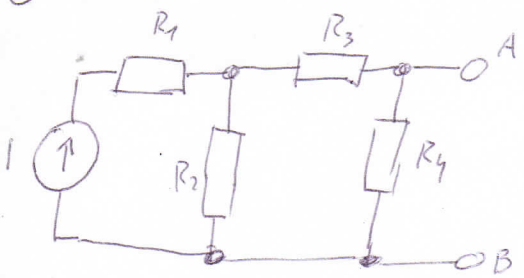
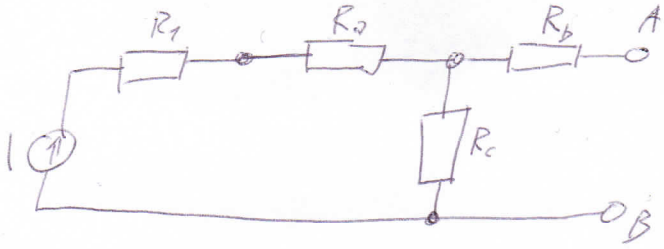
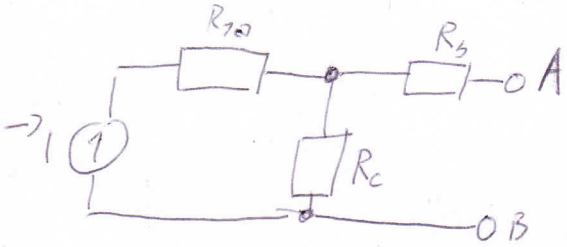
①

Kysela





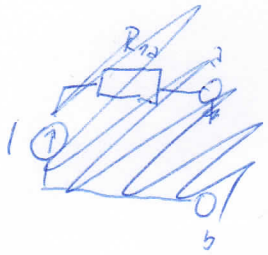


$$R_a = \frac{R_2 \cdot R_3}{R_2+R_3+R_4} = \frac{750 \cdot 1500}{750+1500+9000} = 100\,\Omega$$

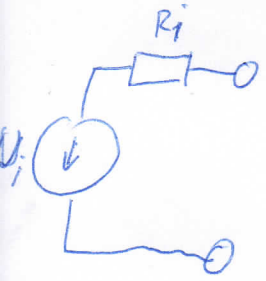
$$R_b = \frac{R_3 \cdot R_4}{R_2+R_3+R_4} = \frac{1500 \cdot 9000}{750+1500+9} = 1200\,\Omega$$

$$R_c = \frac{R_2 \cdot R_4}{R_2+R_3+R_4} = \frac{750 \cdot 9000}{750+1500+9000} = 600\,\Omega$$

$$R_{1a} = R_1 + R_a = 500 + 100 = 600\,\Omega$$



Thévenin



$$U_i = I \cdot R_i$$

$$R_i = R_{1234}$$